

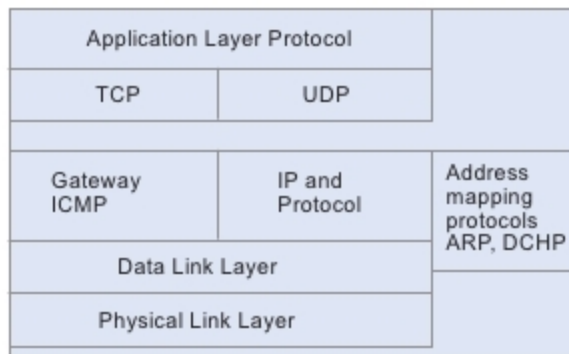


Figure 3: TCP/IP or UDP/IP suite (Internet) model

- normal delivery.
- The session layer manages how to handle packets to be transported by various networks, nodes and links. It establishes a logical session for transport.
- The presentation layer makes the data presentable (encryption/decryption, for example) for transport by syntax, technique for floating point numbers and so on.
- The application layer defines the type of application of data to be transported. It implements, for example, file system operations.

A protocol suite used in the Internet is for the standard networking of isolated networks of various types. It is made of more than one protocol. TCP/IP as well as UDP/IP are both protocol suites developed for the Internet.

TCP/IP is a four-layered protocol (Figure 3), where layer 1 is the network interface service layer. It corresponds to layers 1 and 2 of OSI/ISO protocol. It provides services related to MAC, device drivers, physical medium, physical attachment and physical signals. In this layer, datagrams are packaged into frames.

Layer 2 is the IP layer. This layer corresponds to layer 3 of OSI/ISO protocol. It provides the routing of datagrams as units of data. Logically, these are packets. The service is connectionless; hence, switching is a datagram service rather than virtual circuit switching.

IP provides the basic service of getting the datagram to its destination. IP receives TCP packets from the upper layer TCP and then it forms its own packet, known as IP packets. Each IP packet is associated with several IP

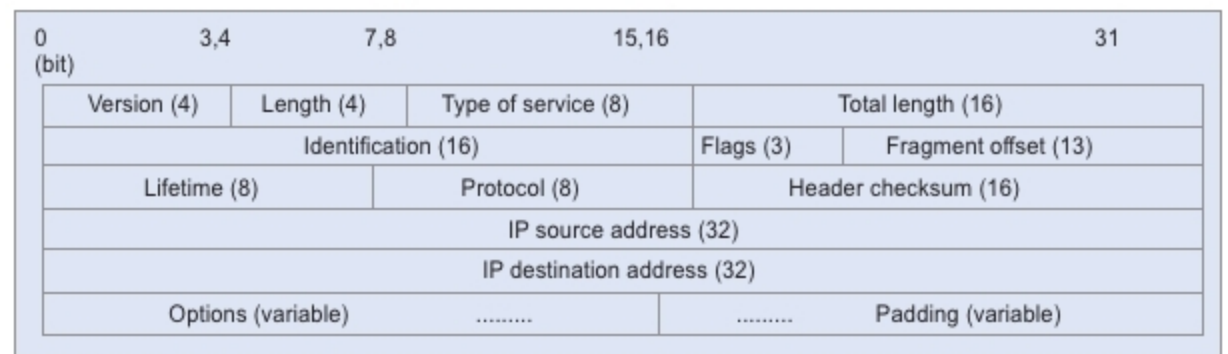


Figure 4: IPv4 headers

headers. For example, original data is fragmented into three packets at TCP layer as: data pack 1, 2 and 3. Each data pack is added with TCP headers to make TCP packets.

TCP Packet = Data Pack + TCP headers (added at TCP layer)

Now, TCP packet is sent to the IP layer. With each of the TCP packets, IP layer adds IP headers making an IP packet (frame or datagram).

IP Packet = TCP packet + IP headers (added by IP layer)

Then, IP datagram is released to the physical layer.

IP header descriptions

The current version of IP that is in use mostly is version 4, denoted as IPv4 (though IPv6 is also now in use in many places). Formation of packets under IPv4 is made with headers as shown in Figure 4. Various headers are described next. Headers at each level are made up of 4 bytes or 32 bits.

Headers at the first level are:

- 4-bit version field, which indicates IP version used in the IP datagram. The most widely-used version is IPv4. IPv6 (IP version 6) has already emerged as a better alternative.
- 4-bit length field, which is also known as the Internet header length (IHL). It contains information about the number of 32-bit words in the header of the IP datagram. The minimum number of headers in any IP datagram is five 32-bit or 20-byte words. This happens when there is no other option or padding field.

In this case, the IHL field contains 0101. As the IHL field can have the maximum value as 1111 (=15), maximum number of headers in any

IP datagram is 15 32-bit or 60-byte words. As the number of headers are in multiples of 32-bit or 4-byte words, headers that an IP datagram can carry are one of the following sizes: 20, 24, 28, 32, 36, 40, 44, 48, 52, 56 or 60.

- 8-bit type of service (TOS) field, which indicates whether the IP datagram should receive any special type of service. Based on the indication about special service offers, priority or precedence is given to the particular packet(s) over other packet(s). This enables the routers to choose the appropriate path for the transport of the packet(s).

In the past, this field was largely unused. TOS field splits into two sub-fields: 5-bit D/T/R/C/R (low delay/high throughput/high reliability/low cost/reserved) and 3-bit precedence.

Precedence is absolute and not relative. An IP datagram for a regular packet with no special priority will have D/T/R/C/R sub-field set as 0000 and precedence sub-field set as 000. Consider a multimedia packet that may need a higher throughput and highest-level precedence. For a multimedia packet like that, IP datagram will set D/T/C/R sub-field as 0100 and precedence sub-field as 111.

TOS field is used by routers to choose the routing path. If a regular datagram that needs a normal service reaches a router, the router selects a normal route as per the outing strategy. But when a datagram requiring minimum cost reaches a router, the latter selects a path with minimum number of hops.

Similarly, for a datagram with higher reliability, the router selects a more reliable (lower BER) path.